Dataset Cards

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Dataset Cards before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Leakage

Without opening the old lesson, explain one key idea from Leakage. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Label Quality

Without opening the old lesson, explain one key idea from Label Quality. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Classification or Regression?

Without opening the old lesson, explain one key idea from Classification or Regression?. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Dataset Cards trace and record the first step you can explain without help.

Use W2: Interpret the audit counts for Dataset inspection and cards. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Dataset Cards explanation against the supplied facts before discussing it.

Use the worked case for Dataset inspection and cards. Proposed explanation: “A complete card certifies data quality”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Changed independent case

Write key dataset-card entries for20 photos: five plants, four views each, one phone, one indoor setting. Growth labels have not been supplied.

> **My independent explanation, trace and limits**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Dataset Cards target to a trace and a justified conclusion.

Write five dataset-card fields for a leaf dataset and explain why '1000 images' alone is insufficient.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Dataset Cards case and show the deciding evidence.

A **dataset card** lists fields but omits source and intended use.

Add information needed to assess suitability.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

A **dataset card** lists 500 images but omits collection conditions and intended use. Add two essential entries and explain their value.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 9 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Dataset Cards adds to Responsible Plant Classifier and keep the evidence.

Publish the Plant Classifier **dataset card** before model experiments begin; update it when data changes.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Write five dataset-card fields for a leaf dataset and explain why '1000 images' alone is insufficient.

> **My independently reasoned answer**

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Define the row and source

> Define the row and source. Apply the supplied Dataset Cards facts and keep your reasoning.

For Robot practice v1, state what one row is intended to represent, the source, the unit and the meaning of None.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Interpret the audit counts

> Interpret the audit counts. Apply the supplied Dataset Cards facts and keep your reasoning.

Explain each number in 5 4 2 1. How many extra ID occurrences are there, and how many rows belong to a repeated-ID group?

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Reject an unsupported correction

> Reject an unsupported correction. Apply the supplied Dataset Cards facts and keep your reasoning.

A learner changes None to 0 and -2 to 2, then reports a clean dataset. Explain why this is not a justified repair and what record should be kept.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Inspect and document independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Inspect and document independently. Apply the supplied Dataset Cards facts and keep your reasoning.

Rules supplied for this independent case

Schema for this exercise: one intended row per distinct trip ID; each row is an (ID, duration) pair. Duration is numeric minutes or None for unmeasured. Negative durations are invalid; a recorded zero is allowed. Count extra ID occurrences beyond the first and do not silently alter the raw data.

A new fictional indoor trip export has (P,4), (Q,None), (Q,None), (R,-1), (S,6), (T,0), with durations in minutes and the same **schema**.

Count raw rows, distinct IDs, missing durations, negative durations and extra ID occurrences. Identify overlap. Draft a short **dataset card** stating source, row meaning, unit, findings, unchanged version and one excluded use.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 State an evidence gap

> State an evidence gap. Apply the supplied Dataset Cards facts and keep your reasoning.

Someone duplicates each row ten times and claims the export now covers more kinds of trips. Explain what changed and what did not.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Dataset Cards

> Build a dataset card that reveals what the photo count hides. Explain the labelled step before

> checking the changed condition.

Build a **dataset card** that reveals what the photo count hides.

Teacher-created dataset: 18 photos of 6 fictional plants, 3 views each, one tablet, indoor fluorescent light. Label rule is visible spot/no visible spot/review.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/9

Dataset Cards

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Describe source

Record teacher-created fictional source, one tablet and indoor fluorescent lighting.





2. Describe units

18 photo rows come from only 6 plant identities.





3. Record limits

Split by plant; state that outdoor use and disease diagnosis have not been evaluated.

CHANGE ONE THING

The dataset adds 6 outdoor photos of the same plants.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

What two counts should the revised card show? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and **limitation** of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Responsible Plant Classifier record.

Project: Responsible Plant Classifier

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and **limitation** of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Build a **dataset card** that reveals what the photo count hides.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Split by plant; state that outdoor use and disease diagnosis have not been evaluated.

1. Record limits

2. Describe units

3. Describe source

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Record limits Split by plant; state that outdoor use and disease diagnosis have not been evaluated. Read the picture in order.

Split by plant; state that outdoor use and disease diagnosis have not been evaluated. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The dataset adds 6 outdoor photos of the same plants. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. Photo count becomes 24, plant count remains 6; lighting coverage changes.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Photo count becomes 24, plant count remains 6; lighting coverage changes. Additional views broaden conditions without creating new independent plant identities.

Repair a missing building block

Review Leakage (ai-4/8). Goal: Choose a split unit that matches the intended **prediction** claim. Short practice: How many independently new plant identities are in the original test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Label Quality (ai-4/6). Goal: Write an observable label rule with boundaries and a review option. Short practice: Why should two annotators not vote away Z's missing **evidence**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Classification or Regression? (ai-4/3). Goal: Identify a task from the meaning of its target. Short practice: Did regression MAE improve when the boundary changed?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

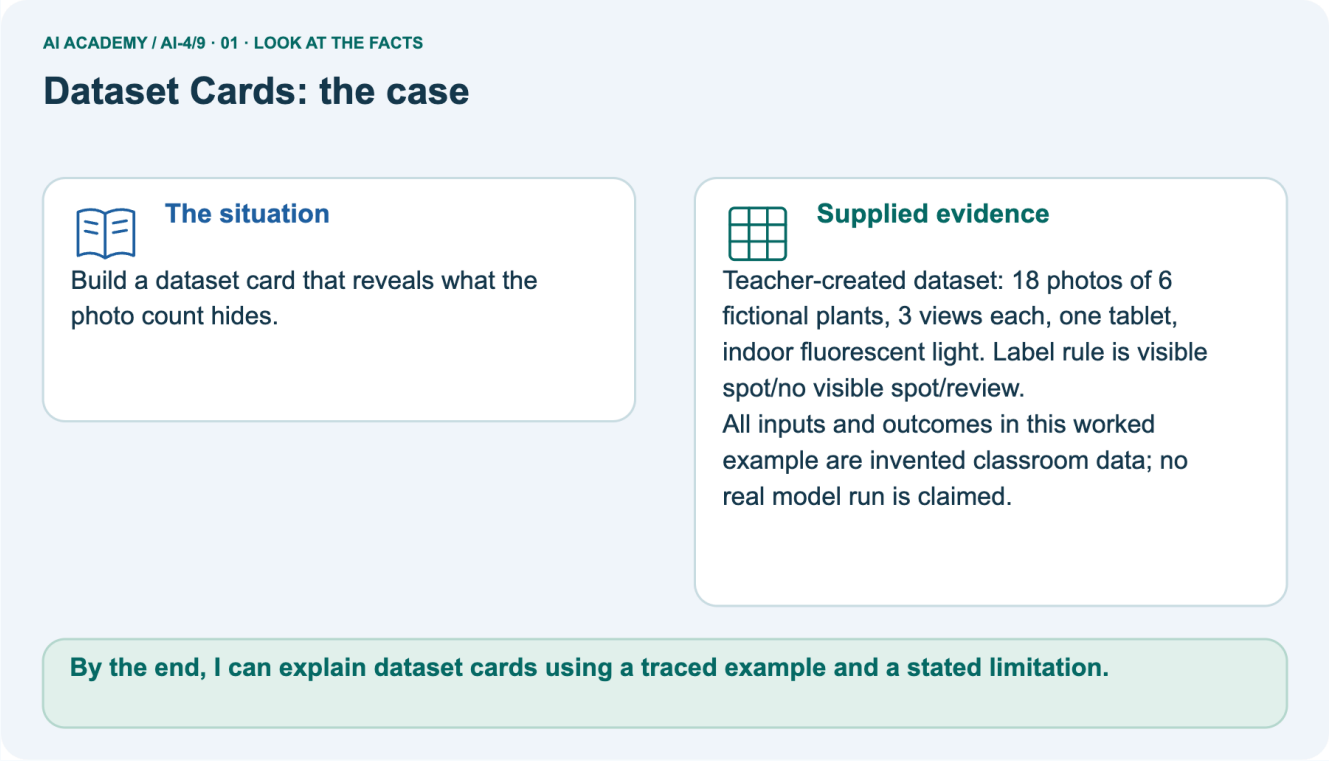
What I repaired, the evidence I kept and the next test

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Dataset Cards picture from its labels, then record a relevant limit.

01. Dataset Cards: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

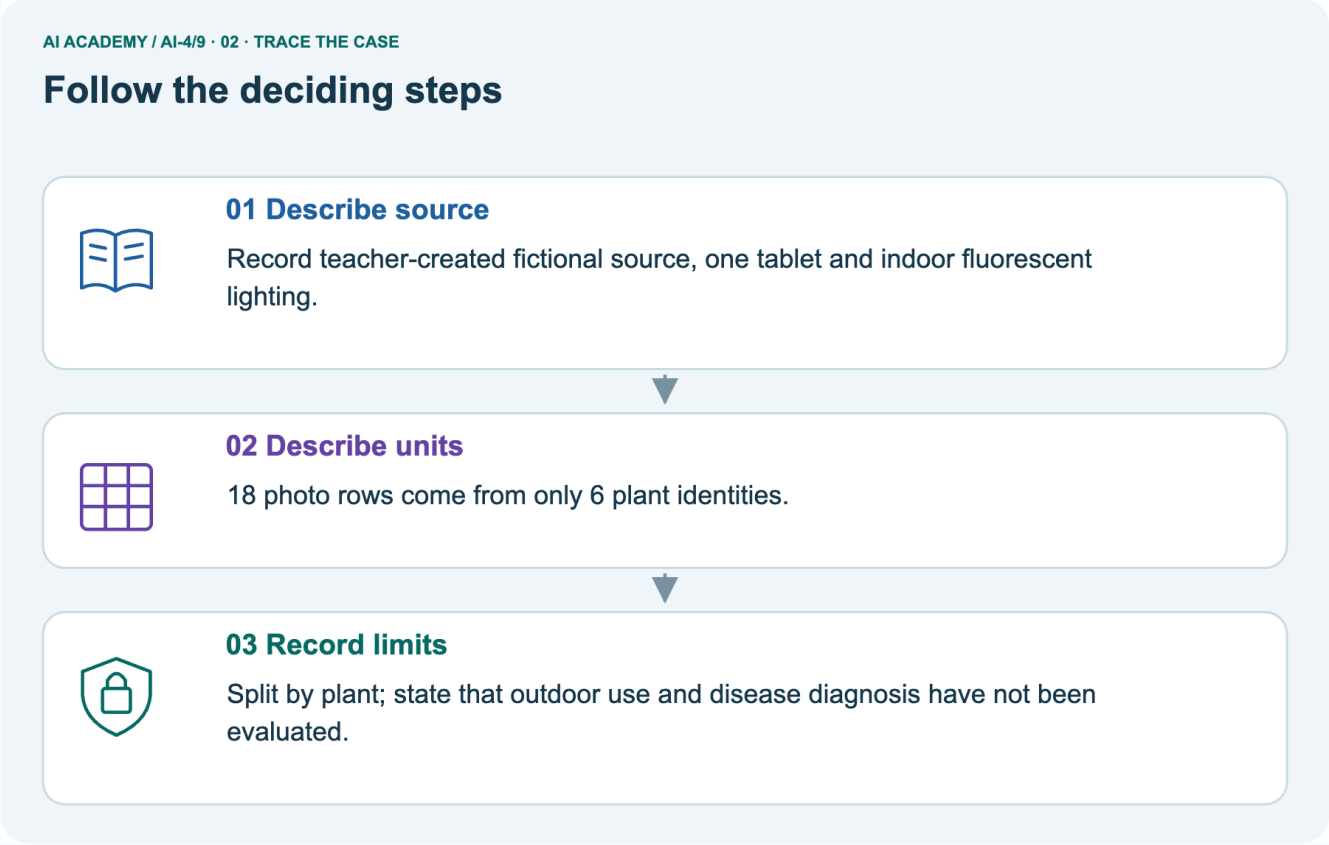
Build a **dataset card** that reveals what the photo count hides. Supplied facts: Teacher-created dataset: 18 photos of 6 fictional plants, 3 views each, one tablet, indoor fluorescent light.

Label rule is visible spot/no visible spot/review. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



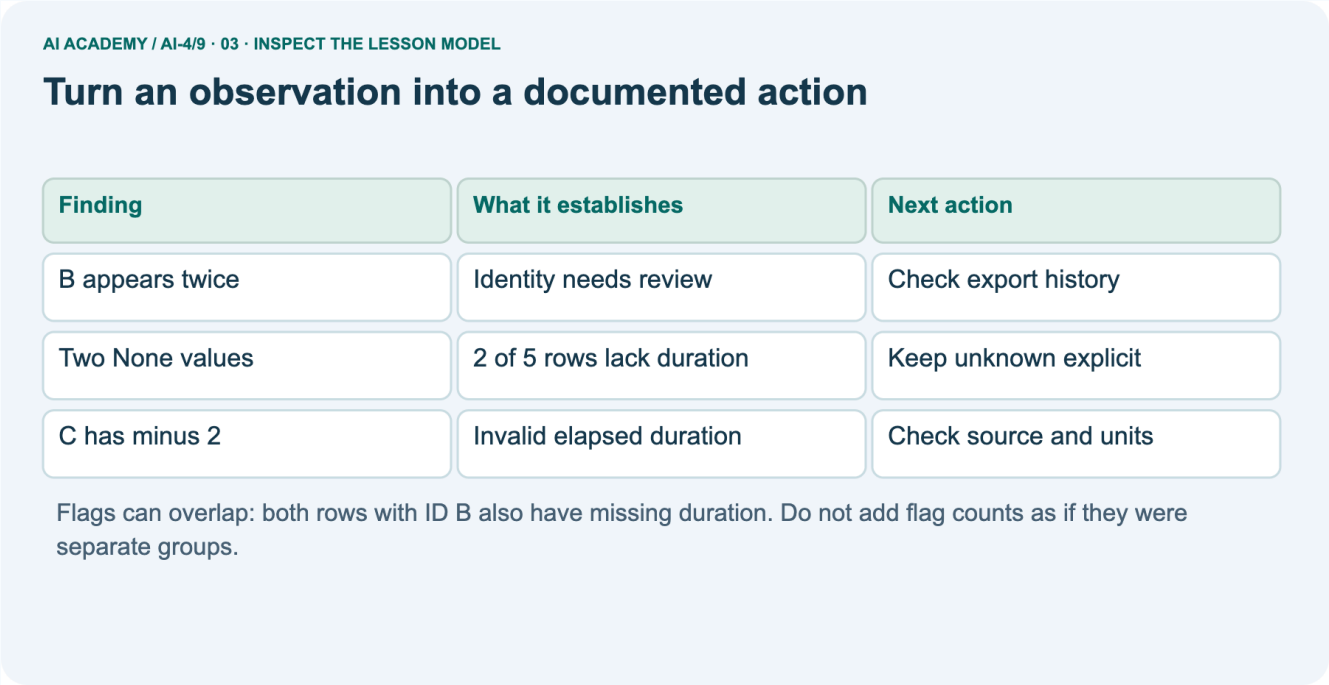
Follow the labelled picture with the matching explanation. Open the vector diagram

Describe source: Record teacher-created fictional source, one tablet and indoor fluorescent lighting. → Describe units: 18 photo rows come from only 6 plant identities. → Record limits: Split by plant; state that outdoor use and disease diagnosis have not been evaluated.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Turn an observation into a documented action



Follow the labelled picture with the matching explanation. Open the vector diagram

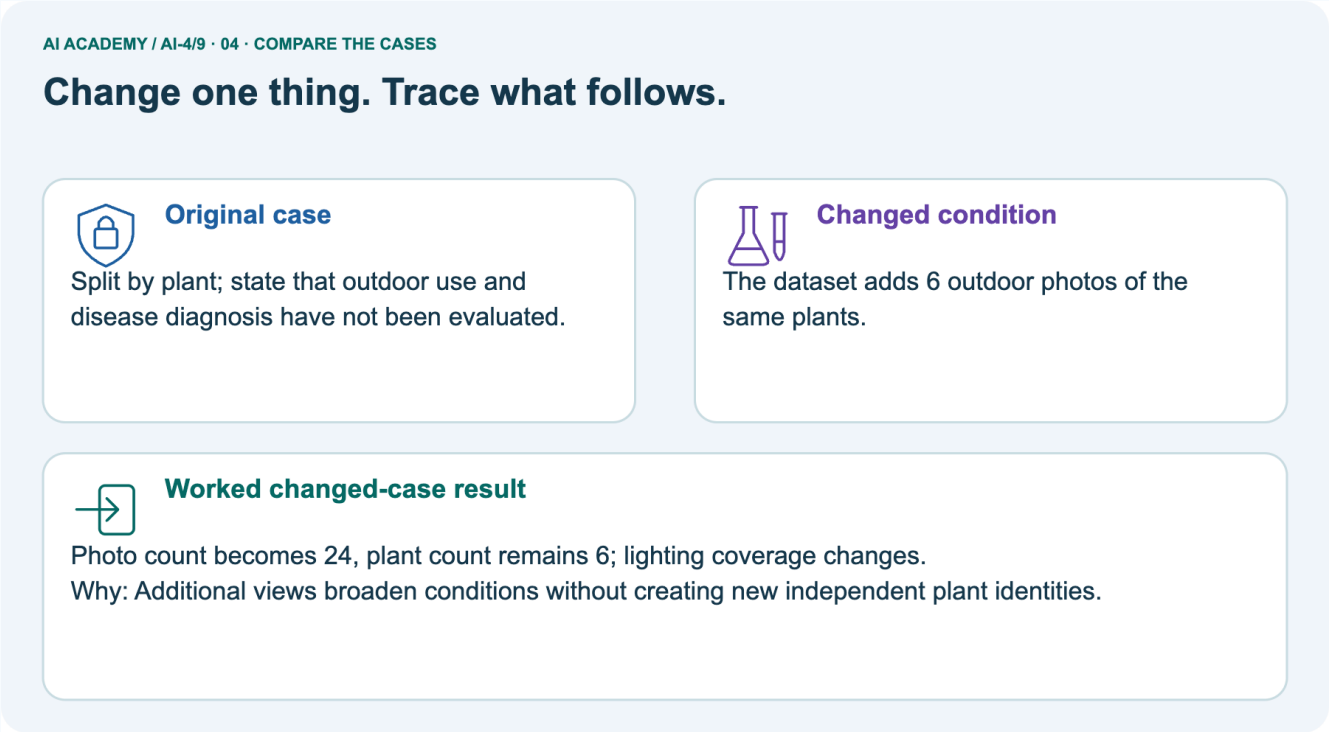
Turn an **observation** into a documented action. B appears twice | Identity needs review | Check export history. Two None values | 2 of 5 rows lack duration | Keep unknown explicit.

C has minus 2 | Invalid elapsed duration | Check source and units

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

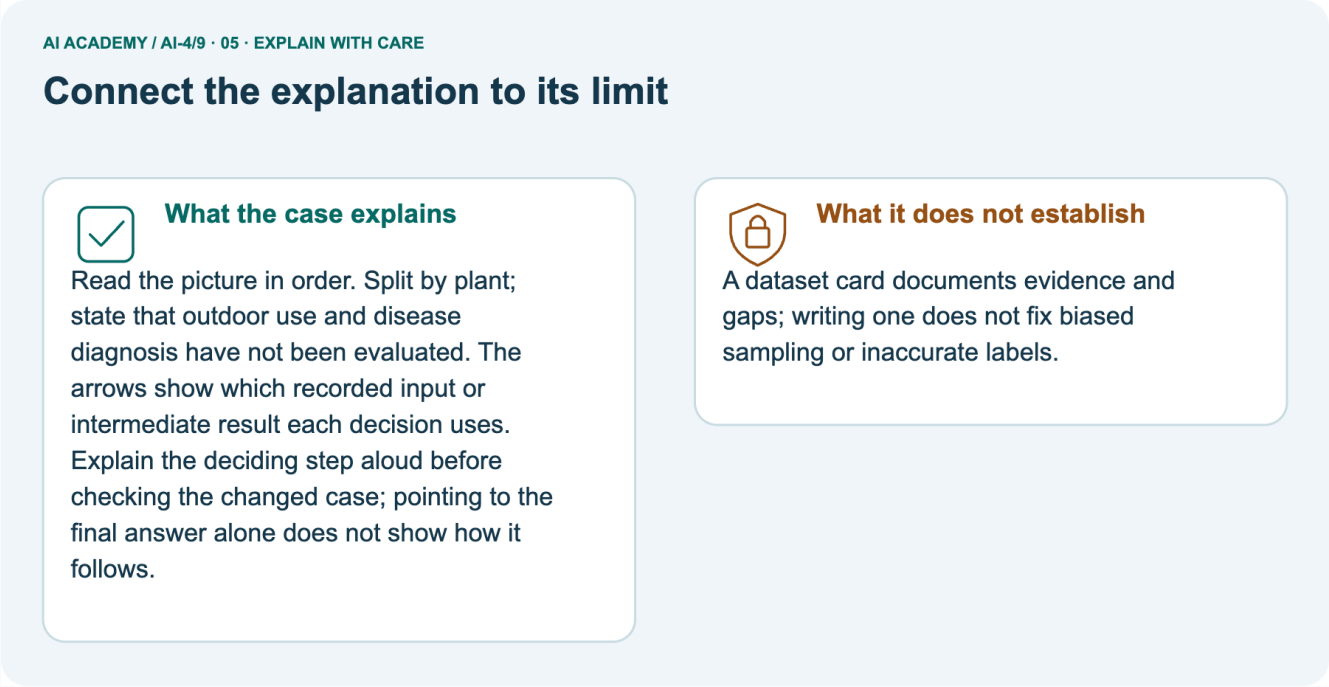
Original case: Split by plant; state that outdoor use and disease diagnosis have not been evaluated.
Changed condition: The dataset adds 6 outdoor photos of the same plants.

Expected result: Photo count becomes 24, plant count remains 6; lighting coverage changes.
Reason: Additional views broaden conditions without creating new independent plant identities.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Split by plant; state that outdoor use and disease diagnosis have not been evaluated. The arrows show which recorded **input** or intermediate result each decision uses.

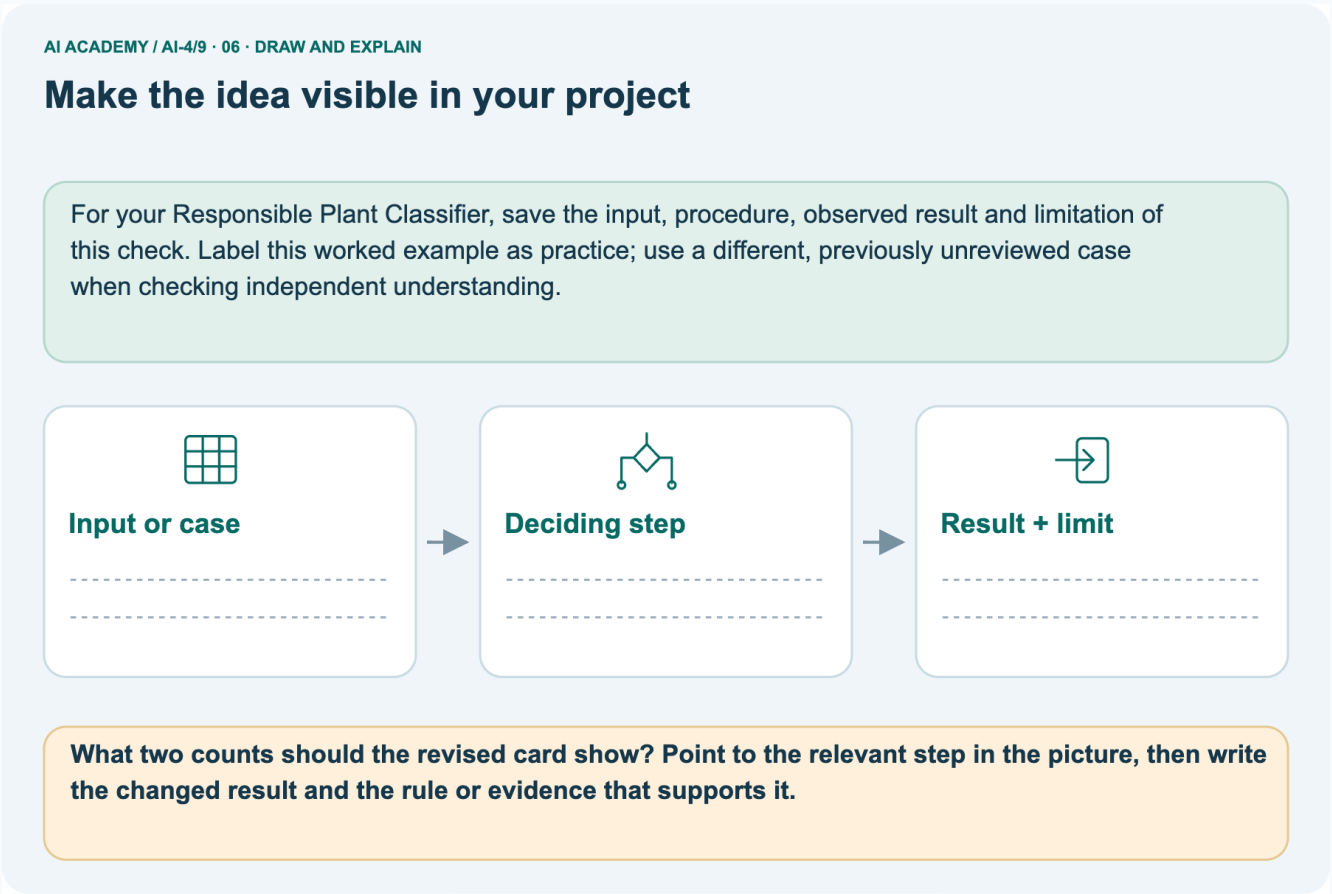
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A **dataset card** documents **evidence** and gaps; writing one does not fix biased sampling or inaccurate labels.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and **limitation** of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What two counts should the revised card show? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.